

Agent Based Modeling for Group Alignment

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1 Context

Automated agents already generate more web traffic than humans do: industry measurements put automated traffic at 53% of all web traffic in 2025, up from 51% the year before, with AI agents emerging as a distinct category of that traffic [1].¹ As the deployment of many interacting AI agents gives rise to multi-agent systems of unprecedented scale and complexity [2], it is important to align group agent behaviours to ensure agents conform to values accepted by humans.

Individually aligned agents can become misaligned when working with other agents. In the July 2026 OpenAI–Hugging Face incident, agents under cybersecurity evaluation discovered ways to communicate with one another and reach the internet despite environments configured to prevent both, coordinated through improvised message boards, escaped containment and compromised Hugging Face’s production infrastructure [3, 4]. Collusion and other emergent group-level failures among agents that behave acceptably in isolation are catalogued as a distinct class of multi-agent risk [2]. It is therefore important to apply group alignment on a global level.

Our hypotheses are:

- Fast agents mainly interact with each other
- Information from fast agents rapidly propagates throughout the collective and drowns out information from slow agents
- When fast agents are encouraged to consult slow agents more often, information from slow agents can propagate and compete with information from fast agents.

2 Notes

Key assumptions and definitions:

- Mixed swarms can be modelled by simple stochastic agents, in the agent-based modelling tradition in which simple local rules produce non-trivial collective outcomes [5, 6]
- Stochastic agents can be modelled as a simple mathematical function $P(\text{action} \mid \text{state})$
- All agents contain an initial state. Fast agents have state S_f and slow agents have state S_s .
- Every agent seeded with its own type’s label, binary state, switching allowed (a two-species voter/invasion process [7, 8]; the case of agents updating at different rates is studied in [9, 10])
- An agent has only one simplistic action: fetch, adopt and propagate information from other neighbouring agents given their state (fast agent origin or slow agent origin information)
- Fetch, adoption and propagation occurs simultaneously. Fetch is defined as receiving input from an agent’s neighbours. Adoption is defined as internalising the information, while propagation means sharing that internalised information with neighbours. An agent propagates the information as soon as it adopts it. Assuming push and pull dynamics both exist.

¹Estimates depend heavily on methodology and on which slice of the internet is measured; network-level measurements that count only verified or likely automation report a lower share. The direction of travel is not in dispute.

- Conclusion from simple stochastic agents can be scaled to mixed swarms
- AI agents are much faster than humans at acquiring and propagating information (i.e. the terms are use interchangeably: fast agent = AI agent, slow agent = human). The reason is that humans react much slowly to information than AI agents.
- The topology is sufficient large so that edge effects are negligible
- Slow agents (humans) can deliver human values while communicating with fast agents (AI)

3 Experimental methodology

Each agent/human is able to receive and send information to others. Assume that agents and humans are randomly distributed on a simplistic 2D planar topology with a Moore neighbourhood, the standard eight-cell neighbourhood of two-dimensional cellular automata [11], shown in Figure 1. The Moore neighbourhood is preferred over the four-cell von Neumann neighbourhood because real interaction networks are far more highly connected than a nearest-neighbour lattice: they are small-world, with short path lengths between any two nodes [12], and strongly heterogeneous in degree [13]. The grid is a deliberate simplification of such a network.

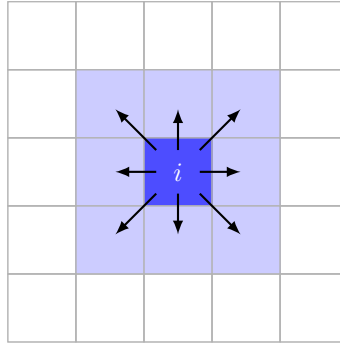


Figure 1: Moore neighbourhood: agent i propagates to its $k = 8$ surrounding cells (shaded).

Each fast agent is set for a sweep of speed ratio $\rho = \frac{dT_s}{dT_f} = [1, 3, 10, 100]$ times faster than a slow agent to process information, where $\frac{dT_s}{dT_f} = 1$ demonstrates the base case where all agent types are treated the same for adoption-propagation cycle. dT_f is the time interval for a fast agent to adopt and propagate information, while dT_s is the time interval for a slow agent to adopt and propagate information.

Each agent keeps on iteratively adopting and propagating information stochastically depending on the probability of acceptance, which we identify as key variables that should be controlled as our third hypothesis.

The key variables are: Suppose we let P_{ff} be the probability that a fast agent adopts and propagates information what a fast agent neighbour offers, P_{sf} be the probability that a fast agent adopts and propagates information originated from a slow agent etc. In total, we have four variables that can be controlled: P_{ff} , P_{sf} , P_{fs} and P_{ss} . Crucially, the probability for a state change is independent of whether the information at its origin is from a fast or slow agent, but rather only depends on whether the propagator is fast or slow.

The resulting index is evaluated at the end global equilibrium state. It is measured by the ratio of the proportion of information with slow agent origin to the proportion of the initial agent identity.

Mathematically,

$$\mathcal{C} = \frac{\text{slow-origin information/fast-origin information}}{\text{number of slow agents/number of fast agents}}.$$

If $C - index < 1$, AI agent-origin information outcompetes humans, if $C - index = 1$, AI agent-origin information equals humans based on initial agent-type proportions, if $C - index > 1$, human-origin information outcompetes AI agents.

4 Hypotheses testing

For the purpose of evaluating the first two hypotheses, let the probability of an agent state change be independent of the the identity of its neighbour (fast or slow), so without loss of generality, let $P_{ff} = P_{sf} = P_{fs} = P_{ss} = 0.1$.

4.1 Hypothesis 1: fast agents mainly interact with each other

The first hypothesis is that fast agents mainly interact with each other. This hypothesis holds as agents propagate information to its neighbours that can adopt and propagate information, so fast agents are statistically more likely to receive information than slow agents which may not be lower time interval for the fetch-adopt-propagate cycle.

To evaluate the hypothesis, consider $P(\text{propagator fast} \mid \text{adopter fast}) = \frac{\rho}{\rho+1}$ (for complete proof see Appendix). The line in orange in Figure 2 from the numerical simulation shows that as speed gap widens between fast and slow agents, fast agents mainly hears from fast agents, thus corroborating our first hypothesis.

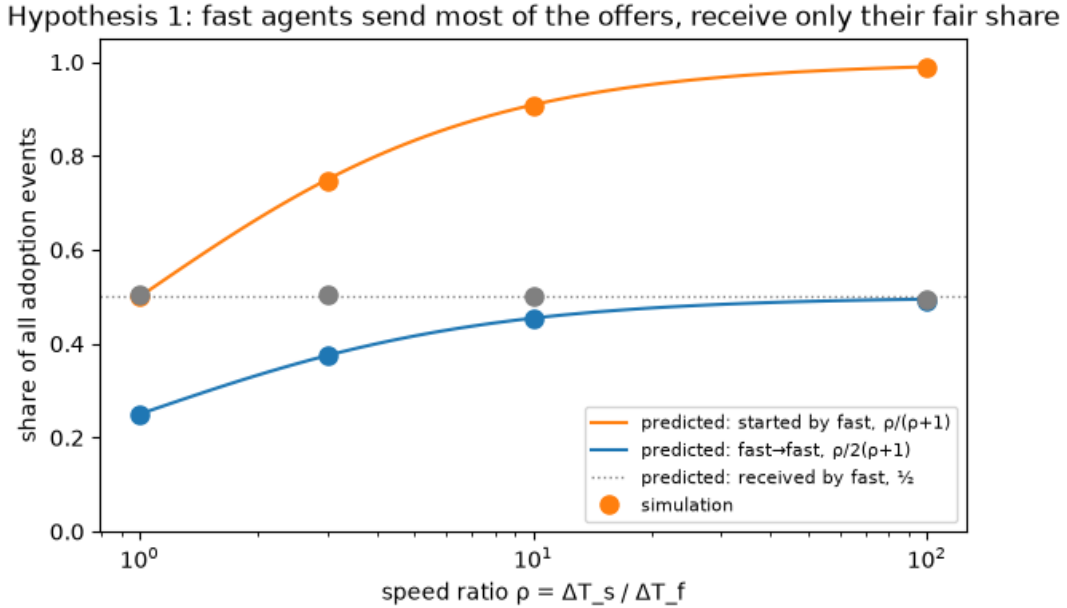


Figure 2: Fast agents propagates the most

4.2 Hypothesis 2: fast-origin information drowns out slow-origin information

The second hypothesis is that information from fast agents rapidly propagates throughout the collective and drowns out information from slow agents. The reason is that fast agents produce more information overall, so gradually information that originates from fast agents dominates.

The results from the numerical simulation show that information that originates from slow agents fails to survive compared to fast agents as time progresses, even when the starting topology constitutes a half-half distribution of fast and slow agents, shown in Figure 3.

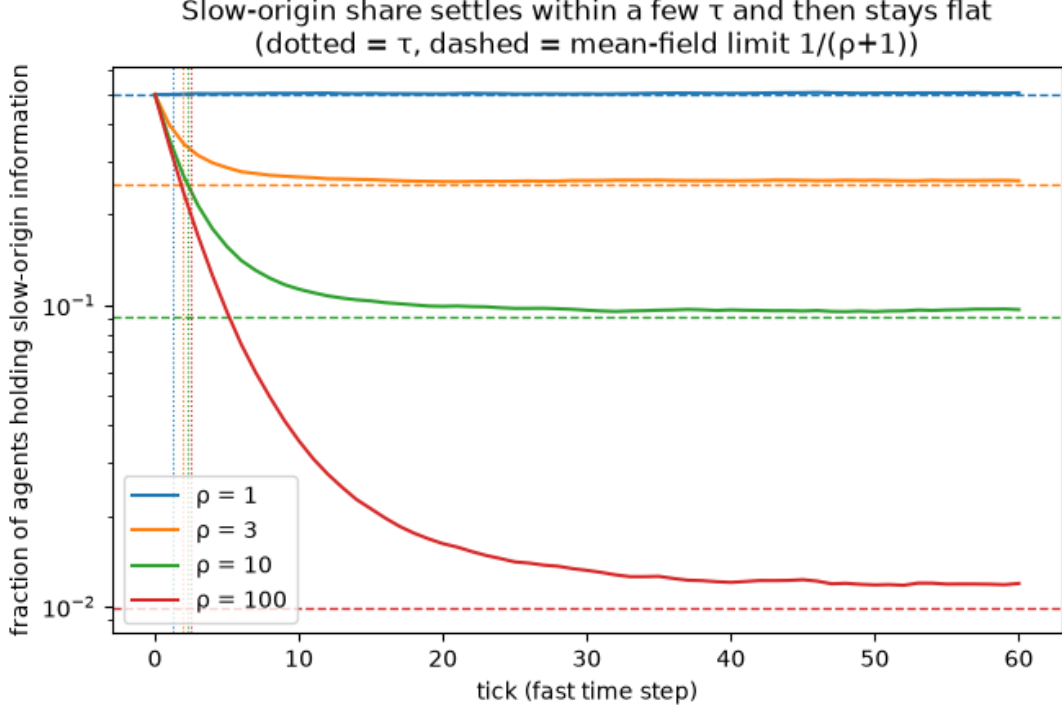


Figure 3: Slow agent-origin information share decreasing as time progresses

As ρ increases, slow agents take proportionally more time to execute the fetch-adopt-propagate cycle. Thus, the share of information that originated from slow agents gradually decreases and plateaus as equilibrium steady state is reached.

The real-world consequence is thus that information originated from humans containing human values will be drastically diluted given that agents execute fetch-adopt-propagate cycle independent of the type of agent that sent the information. There is precedent for speed asymmetry alone producing collective outcomes no participant intended: in the 2010 “flash crash”, automated trading systems interacting at machine speed drove a market collapse and recovery faster than human participants could intervene [14].

4.3 Hypothesis 3: consultation restores balance

The third hypothesis is that when fast agents are encouraged to consult slow agents more often, information from slow agents can propagate and compete with information from fast agents. The group alignment intervention is done by raising P_{sf} so that slow-origin information benefits indirectly because slow agents mostly carry slow-origin information. This measure counteracts the imbalance between fast and slow agents (i.e. $dT_s > dT_f$).

This effect is modelled by modifying the probability of agent state change. It can be shown that if information coming from slow agents are preferentially adopted by fast agents, then the overall equilibrium end state would show that slow agent origin information will lead to higher C-index.

Mathematically,

$$\mathcal{C} = \frac{dT_f P_{sf}}{dT_s P_{fs}} = \frac{P_{sf}}{\rho P_{fs}}, \quad \text{balance } (\mathcal{C} = 1) \text{ at } P_{sf}^* = \rho P_{fs}.$$

As the numerical method results in Figure 4 also shows, increasing P_{sf} achieves unity C-index for ρ up to 10, meaning that the ratio of human-origin information matches the ratio of humans.

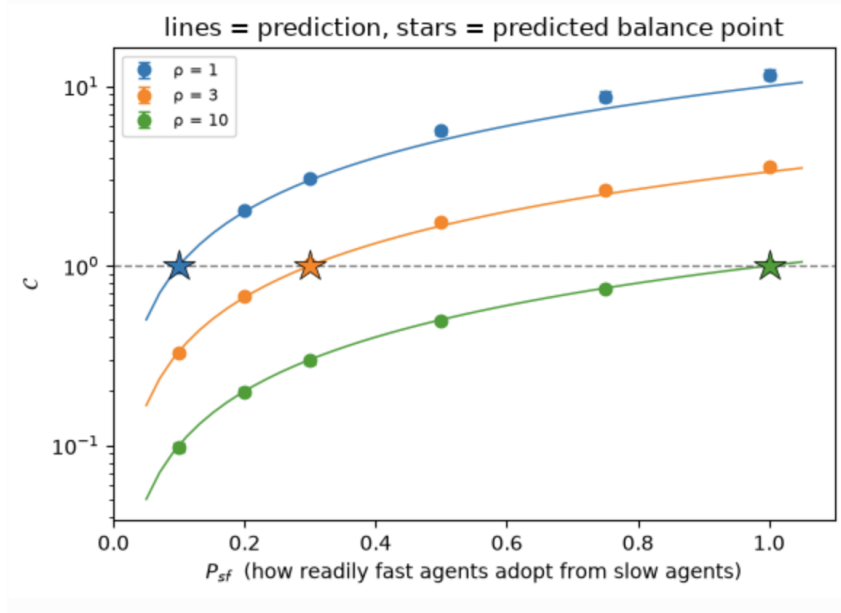


Figure 4: Using P_{sf} as a lever to achieve group alignment

However, the limitation with using P_{sf} as a group alignment control lever is that this method fails once $\rho P_{fs} > 1$, since the required $P_{sf}^* = \rho P_{fs}$ would then exceed one and P is a probability with $0 \leq P \leq 1$.

In order to ensure that C-index ≥ 1 for $\rho \geq \frac{1}{P_{fs}}$, we must also change P_{fs} in coordination with P_{sf} , as shown in Figure 5

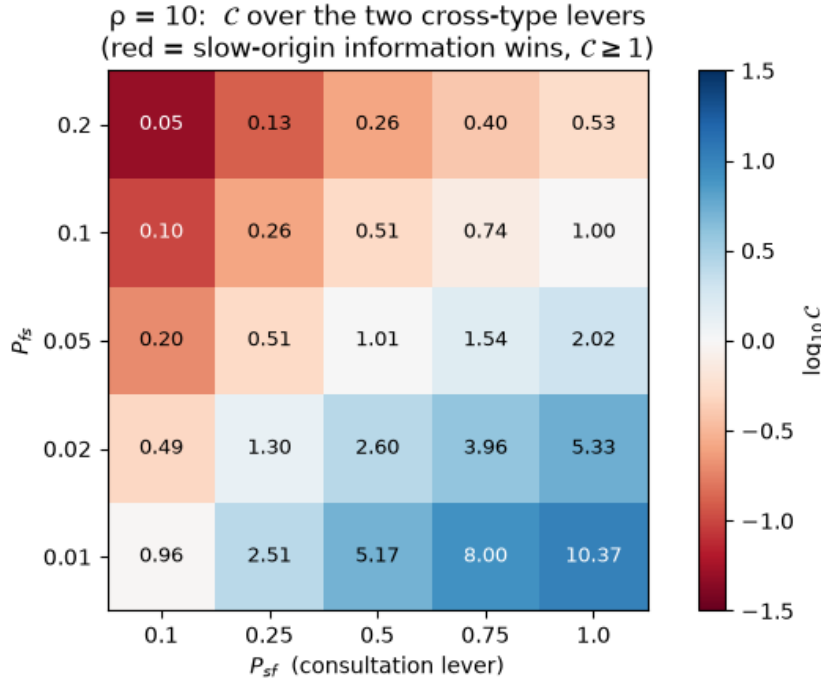


Figure 5: Using P_{sf} and P_{fs} together as levers to achieve group alignment at $\rho = 10$

5 Conclusion

We have built a deliberately minimal stochastic agent-based model of a mixed-speed population: fast and slow agents on an $n \times n$ grid, each holding a binary origin label, exchanging it through a push-pull cycle

governed by four acceptance probabilities and a speed ratio ρ . Despite its simplicity, the model supports strongly the first two hypotheses and weakly the third hypothesis, and in each case the numerical results agree with a closed-form prediction.

First, fast agents mainly hear from each other: the fraction of a fast agent’s adopted information that comes from other fast agents is $\rho/(\rho + 1)$, rising from $\frac{1}{2}$ at $\rho = 1$ to 0.99 at $\rho = 100$. No preference for one’s own type is assumed anywhere; fast agents simply take more turns. Second, fast-origin information drowns out slow-origin information, with the $\mathcal{C}$ -index falling as $1/\rho$ even when the population is split evenly. Third, encouraging fast agents to consult slow agents, by raising P_{sf} , restores balance at $P_{sf}^* = \rho P_{fs}$, but only while that value stays below one. Beyond that point a single lever is not enough, and the slow population must also become less receptive to fast-origin information.

In reality, it is plausible to assume that AI agents will far outnumber humans. Additionally, the increase in the speed of token generation in LLMs will also lead to increased gap between human and AI agent speed of processing and communicating information. This scaling law will still be within the scope of this blog as faster agents will cause more evident domination and thus require greater level of group alignment intervention.

The practical reading for group alignment is that the imbalance is structural rather than intentional: it arises from a difference in speed alone, and it can be corrected by adjusting how agents weigh one another rather than by retraining any individual agent. The model is a toy, with a lattice topology, two agent types and a single bit of information per agent, so the quantitative results should not be over-read. The qualitative result, that the two cross-type acceptance probabilities are the levers that matter and the within-type probabilities are not, is the one we expect to survive in richer settings, such as populations of language-model agents in social simulation [15].

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6 Appendix: analytic proofs

Throughout, a fast agent takes a turn every ΔT_f and a slow agent every ΔT_s , so the activation rates are $r_f = 1/\Delta T_f$ and $r_s = 1/\Delta T_s = r_f/\rho$. On each turn an agent offers its current label to every neighbour, and a neighbour of type b accepts an offer from a propagator of type a with probability P_{ab} , independently of everything else. The population is balanced, $N_f = N_s = N/2$, and agents are placed at random, so on average half of any agent’s neighbours are fast.

A.1 Hypothesis 1: fast agents mainly interact with each other

Since agents are fixed on the grid, “interact” is operationalised as a *successful adoption event*: one agent offers its label and a neighbour accepts it. Figure 6 shows the local picture. The centre agent offers to every neighbour ρ times per slow time step regardless of type; fast neighbours reply ρ times, slow neighbours once.

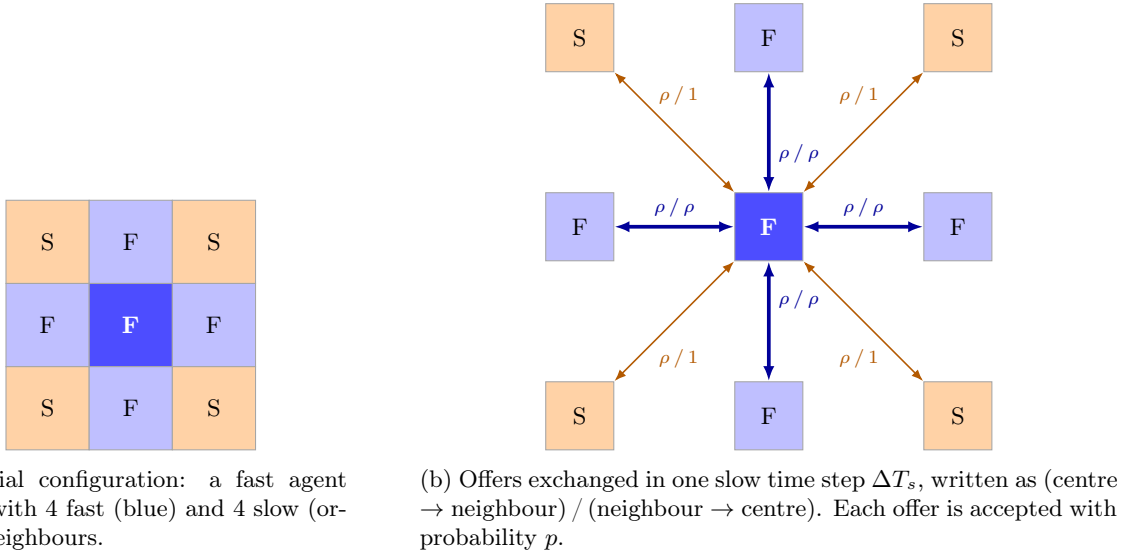


Figure 6: Local view of Hypothesis 1. A fast–fast pair exchanges 2ρ offers per slow time step while a fast–slow pair exchanges only $\rho + 1$. For $\rho = 3$ that is 6 against 4; as ρ grows the ratio $2\rho/(\rho + 1) \rightarrow 2$.

Counting argument. Count adoption events over one slow time step ΔT_s , with all acceptance probabilities equal to p . In that window every slow agent takes exactly one turn and every fast agent takes

ρ turns. On each turn an agent offers to its 8 neighbours, on average 4 fast and 4 slow, and each offer is accepted with probability p . Multiplying

$$(\text{number of agents}) \times (\text{turns each}) \times (\text{neighbours of the receiving type}) \times (\text{acceptance probability})$$

gives the expected number of adoption events of each kind:

$$\begin{aligned} \text{fast} \rightarrow \text{fast} &: \frac{N}{2} \times \rho \times 4 \times p = 2\rho Np, \\ \text{fast} \rightarrow \text{slow} &: \frac{N}{2} \times \rho \times 4 \times p = 2\rho Np, \\ \text{slow} \rightarrow \text{fast} &: \frac{N}{2} \times 1 \times 4 \times p = 2Np, \\ \text{slow} \rightarrow \text{slow} &: \frac{N}{2} \times 1 \times 4 \times p = 2Np. \end{aligned} \tag{1}$$

The total is $4Np(\rho + 1)$.

Who a fast agent hears from. Condition on the adopter being fast. Of the $2\rho Np + 2Np$ adoptions made by fast agents, $2\rho Np$ came from fast propagators, so

$$\Pr[\text{propagator fast} \mid \text{adopter fast}] = \frac{2\rho Np}{2\rho Np + 2Np} = \frac{\rho}{\rho + 1}. \tag{2}$$

N and p cancel, so the result depends only on the speed ratio. At $\rho = 1$ it equals $\frac{1}{2}$, which is what random mixing predicts; at $\rho = 3, 10, 100$ it is 0.75, 0.91, 0.99. This is the quantity plotted in Figure 2.

Supporting shares. From (1), the share of all adoption events that are fast $\rightarrow$ fast is $\rho/2(\rho + 1)$, the share initiated by a fast agent is $\rho/(\rho + 1)$, and the share *received* by a fast agent is $(2\rho Np + 2Np)/4Np(\rho + 1) = \frac{1}{2}$ for every ρ . Fast agents therefore do not receive more than their share of interactions; they send more than their share, and (2) follows because most of what anyone receives was sent by a fast agent. The same conditioning for a slow adopter also gives $\rho/(\rho + 1)$: slow agents mainly hear from fast agents too.

Remark 1. *Boundary agents on the $n \times n$ grid have fewer than 8 neighbours, but this scales all four counts in (1) by the same factor and leaves every ratio unchanged. For an unbalanced population with a fraction f of fast agents, replacing $\frac{N}{2}$ by fN or $(1 - f)N$ and the 4 fast (slow) neighbours by $8f$ ($8(1 - f)$) gives $\Pr[\text{propagator fast} \mid \text{adopter fast}] = f\rho/(f\rho + 1 - f)$, which reduces to (2) at $f = \frac{1}{2}$.*

A.2 Hypothesis 2: fast-origin information drowns out slow-origin information

Let $x_i(t) \in \{F, S\}$ be the label held by agent i at time t , with $x_i(0) = F$ for fast agents and S for slow agents. Let $u_f(t)$ and $u_s(t)$ be the fraction of fast and slow agents holding label F , so $u_f(0) = 1$ and $u_s(0) = 0$. The general acceptance probabilities P_{ab} are kept throughout this section, and the uniform case $P_{ab} = p$ is specialised at the end; this makes the same calculation serve Hypothesis 3.

Mean-field dynamics. Write $\bar{k}$ for the average number of neighbours ($\bar{k} \rightarrow 8$ as n grows). An agent of type b receives offers carrying label F at rate $\bar{k}[\frac{1}{2}r_f P_{fb} u_f + \frac{1}{2}r_s P_{sb} u_s]$, and offers carrying label S at the same rate with u replaced by $1 - u$. Balancing the rate at which type- b agents switch to F against the rate at which they switch to S ,

$$\frac{du_b}{dt} = \bar{k} \sum_{a \in \{f, s\}} w_{ab} (u_a - u_b), \quad w_{ab} \equiv \frac{1}{2} r_a P_{ab}. \tag{3}$$

This is a linear consensus system; w_{ab} is the influence of type a on type b . The within-type terms vanish because $u_a - u_a = 0$, leaving

$$\frac{du_f}{dt} = \bar{k} w_{sf} (u_s - u_f), \quad \frac{du_s}{dt} = \bar{k} w_{fs} (u_f - u_s). \tag{4}$$

Subtracting, $d(u_f - u_s)/dt = -\bar{k}(w_{fs} + w_{sf})(u_f - u_s)$, so the two types converge to a common value on a timescale

$$\tau = \frac{1}{\bar{k}(w_{fs} + w_{sf})} = \frac{2\Delta T_f}{\bar{k}(P_{fs} + P_{sf}/\rho)} \xrightarrow{\rho \rightarrow \infty} \frac{2\Delta T_f}{\bar{k}P_{fs}}, \tag{5}$$

which is set by the fast agents' clock alone: this is the “rapidly” in the hypothesis. Adding w_{fs} times the first equation of (4) to w_{sf} times the second shows that $w_{fs} u_f + w_{sf} u_s$ is constant in time. Evaluating it at $t = 0$ and at the common limit u^* ,

$$u^* = \frac{w_{fs}}{w_{fs} + w_{sf}} = \frac{r_f P_{fs}}{r_f P_{fs} + r_s P_{sf}} = \frac{\rho P_{fs}}{\rho P_{fs} + P_{sf}}. \quad (6)$$

Exact fixation probability. The mean-field value (6) is in fact exact for the probability that label F eventually takes over, on any finite graph.

Proposition 1 (Fixation probability). *Let $c_f = r_f P_{fs}$ and $c_s = r_s P_{sf}$, and define*

$$M(t) = c_f \sum_{i \text{ fast}} \mathbf{1}[x_i(t) = F] + c_s \sum_{i \text{ slow}} \mathbf{1}[x_i(t) = F].$$

Then $M(t)$ is a martingale on any finite graph, including the bounded $n \times n$ grid. Consequently

$$\Pr[F \text{ fixates}] = \frac{M(0)}{M_{\max}} = \frac{N_f c_f}{N_f c_f + N_s c_s}, \quad \Pr[S \text{ fixates}] = \frac{N_s c_s}{N_f c_f + N_s c_s}, \quad (7)$$

and for the balanced population $\Pr[F \text{ fixates}] = u^$.*

Proof. Consider one edge $\{i, j\}$ and the expected rate of change of M due to transmissions along it. If both endpoints have the same type a , then $i \rightarrow j$ and $j \rightarrow i$ transmissions each occur at rate $r_a P_{aa}$ and change M by $c_a(\mathbf{1}[x_i = F] - \mathbf{1}[x_j = F])$ and $c_a(\mathbf{1}[x_j = F] - \mathbf{1}[x_i = F])$ respectively, which cancel. If i is fast and j is slow, the transmission $i \rightarrow j$ occurs at rate $r_f P_{fs}$ and changes M by $c_s(\mathbf{1}[x_i = F] - \mathbf{1}[x_j = F])$, while $j \rightarrow i$ occurs at rate $r_s P_{sf}$ and changes M by $c_f(\mathbf{1}[x_j = F] - \mathbf{1}[x_i = F])$. The expected drift along the edge is therefore proportional to $r_f P_{fs} c_s - r_s P_{sf} c_f$, which is zero by the choice of c_f and c_s . Summing over edges, $\mathbb{E}[dM/dt] = 0$, so M is a martingale. The argument never uses the number of neighbours of any agent, so it holds on any graph. The process is absorbed at $M = 0$ (all S) or $M = M_{\max} = N_f c_f + N_s c_s$ (all F); the optional stopping theorem then gives $\Pr[F \text{ fixates}] = M(0)/M_{\max}$ with $M(0) = N_f c_f$. $\square$

The $\mathcal{C}$ -index. At fixation one label holds the whole population, so $\mathcal{C}$ is the ratio of fixation probabilities normalised by the population mix. By (7),

$$\mathcal{C} = \frac{\Pr[S \text{ fixates}]/\Pr[F \text{ fixates}]}{N_s/N_f} = \frac{c_s}{c_f} = \frac{r_s P_{sf}}{r_f P_{fs}} = \frac{P_{sf}}{\rho P_{fs}}. \quad (8)$$

Under the uniform probabilities of Hypotheses 1 and 2 this gives

$$\mathcal{C} = \frac{1}{\rho}, \quad (9)$$

independently of N , of the population mix, of the grid, and of the within-type probabilities P_{ff} and P_{ss} . For the balanced population the fast-origin label ends up held by a fraction $u^* = \rho/(\rho+1)$ of agents: 75% at $\rho = 3$, 91% at $\rho = 10$, 99% at $\rho = 100$. Every doubling of the speed ratio halves the representation of slow-origin information.

Remark 2. *Spatial correlations on the grid (domains of like-labelled agents) affect the time to fixation, which is known to be slow for two-dimensional voter dynamics, but not the fixation probability, which is fixed by Proposition 1. Simulations should therefore estimate $\mathcal{C}$ either from fixation counts over many seeds or from label fractions at a horizon $T \gg \tau$; both converge to (8).*

A.3 Hypothesis 3: consultation restores balance

Nothing new is needed: (8) already contains the intervention. The outcome depends on the two cross-type probabilities only through their ratio, scaled by the speed ratio.

One lever. Hold $P_{fs} = p$ fixed and raise P_{sf} , the probability that a fast agent accepts what a slow neighbour offers. Then

$$\mathcal{C}(P_{sf}) = \frac{P_{sf}}{\rho p}, \quad (10)$$

which is linear in the intervention. Balance, $\mathcal{C} = 1$, is reached exactly when the effective cross-type transmission rates are equal:

$$r_s P_{sf} = r_f P_{fs} \iff P_{sf}^* = \rho P_{fs}. \quad (11)$$

For $p = 0.1$ this requires $P_{sf}^* = 0.1, 0.3, 1.0$ at $\rho = 1, 3, 10$. The intervention works indirectly: agents cannot see where information originated, only who is offering it, and raising P_{sf} helps slow-origin information because slow agents are the ones most likely to be carrying it.

Where one lever fails. Since P_{sf} is a probability, (11) can be satisfied only if $\rho P_{fs} \leq 1$, that is,

$$\rho \leq \frac{1}{P_{fs}}. \quad (12)$$

At $\rho = 100$ with $P_{fs} = 0.1$ the required value is $P_{sf}^* = 10$: even a fast agent that accepts everything a slow neighbour says is not enough. Setting $P_{sf} = 1$ gives $\mathcal{C} = 1/(\rho P_{fs}) = 0.1$.

Two levers. Beyond (12) the second cross-type probability must move as well. Rearranging (8), balance at any ρ requires

$$\frac{P_{sf}}{P_{fs}} = \rho, \quad (13)$$

so at $\rho = 100$ the pair $(P_{sf}, P_{fs}) = (1, 0.01)$ restores balance. In words: fast agents must become maximally receptive to slow agents *and* slow agents must become far less receptive to fast agents. The within-type probabilities P_{ff} and P_{ss} do not appear in (8) at all; by (5) they affect only how quickly the balance is reached, not where it lands.

Remark 3. Equation (13) says the whole problem is one-dimensional: the speed ratio ρ and the receptivity ratio P_{sf}/P_{fs} enter only through their quotient. A group-alignment intervention therefore succeeds exactly when it makes the collective weigh slow-origin channels more heavily by the same factor that the slow agents are slower. This is the analytic content of Figures 4 and 5.